TripNest AI: Requirement Traceability

Version 2.0 · 2 November 2026. Chain: business requirement → product requirement → epic → story → source files → API → database → tests. Test names are checked automatically against the test files by `docs/build_pdfs.py`.

1. Business requirement to epic

BR	Product requirements	Epic	Stories
BR-001 Predict crowds for a date and time	FR-004, FR-006, FR-007, FR-010, FR-012, AI-001, AI-002	EPIC-001	STORY-001 to STORY-007, STORY-020
BR-002 Recommend by predicted crowd and preference	FR-002, FR-003, FR-005, FR-008, FR-009, AI-003	EPIC-002	STORY-008 to STORY-015
BR-003 Simple, no registration	FR-001, FR-010, FR-011, NFR-001, NFR-004	EPIC-003	STORY-016 to STORY-021
BR-004 Honest about data	FR-012, AI-004, SEC-004	EPIC-001, EPIC-002	STORY-006, STORY-007, STORY-009, STORY-011
BR-005 Public and reproducible	NFR-005, SEC-001 to SEC-003	EPIC-004	STORY-022 to STORY-025

2. Story to implementation and tests

Source paths are under `frontend/assets/js/` (front end) and `backend/app/` (back end). Test names are listed in `USER_STORIES.md` and checked by `docs/build_pdfs.py`.

Story	Source files	API	Database / data	Test files
STORY-001	`destination.js`, `explore.js`	`/api/destinations/{slug}/prediction`	`destinations`, `attractions`	`e2e/test_journeys.py`, `test_core_features.py`
STORY-002	`destination.js`, `services/prediction.py`	same	same	`test_core_features.py`, `e2e`
STORY-003	`destination.js`, `core.js` chart	same	same	`test_core_features.py`
STORY-004	`ui.js` timeline, `explore.js`, `prediction.timeline/best_window`	`/api/explore`, prediction	same	`test_explore.py`, `e2e`
STORY-005	`destination.js`, `prediction.py`	prediction	same	`test_core_features.py`, `e2e`
STORY-006	`prediction.provenance`	prediction, explore	model file	`test_core_features.py`, `e2e`
STORY-007	`services/forecast.py`, `crowd_sim.py`, `scripts/train_model.py`	`/api/model`	`backend/models`	`test_crowd_engine.py`, `test_core_features.py`
STORY-008	`landing.js`, `search.js`, `data/cities.py`	`/api/locations/*`	static city list	`test_core_features.py`, `e2e`
STORY-009	`search.js`, `landing.js`	`/api/locations/nearest`	static city list	`test_explore.py`, `e2e`
STORY-010	`explore.js`, `services/explore.py`	`/api/explore`	`destinations`, `attractions`	`test_explore.py`, `e2e`
STORY-011	`explore.js`, `services/explore.py`	`/api/explore`	same	`test_explore.py`, `e2e`
STORY-012	`explore.js` (refine bar)	`/api/explore`	same	`test_explore.py`, `e2e`
STORY-013	`explore.js`, `ui.js`, `routers/catalog.py`	`/api/explore` (map), `/api/crowd/map`	same	`test_explore.py`, `test_map_photos.py`, `e2e`
STORY-014	`services/recommender.py`	`/api/recommendations`	same	`test_core_features.py`
STORY-015	none in the UI	`/api/crowd/compare`	same	`test_core_features.py`
STORY-016	`landing.js`, `main.js`, `styles.css`	none	none	`e2e`
STORY-017	`destination.js`, `ui.js`	prediction	same	`e2e`
STORY-018	`styles.css`, all pages	none	none	`e2e`
STORY-019	`ui.js` (photo fallback), `main.js`	none	none	`e2e`
STORY-020	`data/catalog*.py`, `scripts/fetch_images.py`, `models.py`, `services/seed.py`, migration `destination photos`	`/api/states`, `/api/destinations`	`states`, `destinations` (photo columns), `attractions`	`test_catalog_crowd.py`, `test_explore.py`, `test_schema_upgrade.py`
STORY-021	`main.js`	none	none	`e2e`
STORY-022	`backend/tests`, `e2e`	none	none	all (`TEST_RESULTS.md`)
STORY-023	`config.py`, `main.py`	`/api/health`	none	`test_core_features.py`

Story	Source files	API	Database / data	Test files
STORY-024	`Dockerfile`, `render.yaml`	`/api/health`	PostgreSQL (production)	none yet: **not deployed**
STORY-025	`docs/*`	none	none	`docs/build_pdfs.py` (page counts, text, test references)

3. Non-functional, AI and security requirements

ID	Requirement	Implemented in	Verified by
NFR-001	Responsive, no sideways scroll	`styles.css`	`test_every_screen_size_supports_the_whole_flow`
NFR-002	Fast results	batched forecasts, cache (`runtime.py`)	measured manually: recommendations 0.02–0.05 s warm (no automated performance test)
NFR-003	Accessibility basics	labels, focus styles, reduced motion	design inspector; no assistive-technology audit
NFR-004	Loading, empty, error states	all pages	STORY-017 tests
NFR-005	Reproducible setup	`README.md`, pinned requirements	STORY-019
AI-001, AI-002	Chronological split; baselines	`forecast.py`	`test_training_uses_chronological_train_validation_test_split`, `test_model_beats_both_baselines_on_held_out_test_period`
AI-003	Recommendation uses predictions	`recommender.py`	`test_busier_dates_lower_the_crowd_score_for_the_same_destination`
AI-004	No invented live data	provenance labels	`test_prediction_states_what_it_is_and_what_it_cannot_do`
AI-005	Real data can be loaded later	`crowd_sim.counts` contract, `train_model.py`	documented; not exercised with real data
SEC-001	Secrets in environment	`config.py`	production start refuses without secrets (manually executed, see TEST_RESULTS)
SEC-002	Input validation	`discover.py`	`test_recommendation_input_validation`, `test_prediction_defaults_and_validation`
SEC-003	Security headers	`main.py`	`test_security_headers_present`
SEC-004	Location privacy	`home.js`	`test_granted_location_is_used_with_a_dynamic_greeting`